Curriculum Vitae

Sangjun Jeon, Ph.D.

Hyundai-NTU-A*STAR Corporate Lab, School of Mechanical and Aerospace Engineering, Nanyang Technological University, #N3.2-B1-02, 65 Nanyang Drive, Singapore, 637460

Phone: +65-8560-8259, +82-10-7130-9354

Email: sangjun.jeon@ntu.edu.sg

KEY MESSAGES

By delving deep into the intersection of materials science and sustainable manufacturing processes, my research aims to drive a paradigm shift in how we conceive, produce, and utilize products in society.

RESEARCH PROFILE

My research focuses on sustainable materials and manufacturing, integrating composite recycling, additive manufacturing, interface engineering, and AI-driven process optimization. I investigate process–structure–property relationships in recycled composites and additively manufactured materials to improve mechanical performance and support material reuse and reliable engineering applications.

RESEARCH INTERESTS

Sustainable Materials and Manufacturing

- **Composite Recycling and Remanufacturing:** Recovery, recycling, and remanufacturing of end-of-life carbon-fiber composites, with additive manufacturing routes for reusing recovered fibers in engineering components.
- **AI-Driven Materials Processing:** Machine learning and active learning for data-efficient process–property prediction, inverse design, and process optimization in recycled composites, metallic materials, and additively manufactured structures, with an emphasis on achieving targeted mechanical performance.
- **Advanced Manufacturing and Interface Engineering:** Additive and hybrid manufacturing, surface modification, and pre-/post-processing to improve interlayer bonding, interfacial strength, and structural integrity in composite, metallic, and polymer–metal systems, supporting the engineering use of recycled and additively manufactured materials.
- **Mechanical Performance and Durability:** Experimental characterization and multiscale analysis of process–structure–property relationships, focusing on fatigue and temperature-dependent behavior to assess the durability and application limits of recycled and additively manufactured materials.

EDUCATION

Aug. 2025	Ph.D. Future Convergence Engineering **[President's Doctoral Award recipient]** Kongju National University, Cheonan, Republic of Korea
Feb. 2022	M.S. Future Convergence Engineering Kongju National University, Cheonan, Republic of Korea
Aug. 2020	B.S. Mechanical and Automotive Engineering **[Valedictorian (top of graduating class), College of Engineering]** Kongju National University, Cheonan, Republic of Korea

PROFESSIONAL APPOINTMENTS

Present	Research Fellow, Hyundai-NTU-A*STAR Corporate Lab Nanyang Technological University, Singapore
Dec. 2025	Lecturer, Kongju National University, Cheonan, Republic of Korea Department of Mechanical and Automotive Engineering (Aug. – Dec. 2025)
May 2025	Visiting Ph.D. Student, Nanyang Technological University, Singapore School of Mechanical and Aerospace Engineering (Apr. 2024 – May 2025, funded by Korea NRF) **Received the NRF President's Award for outstanding overseas research under the BK21 FOUR program.**

RESEARCH PROJECTS AND GRANTS

Aug. 2025	The National University Development Project of the Ministry of Education Role: Principal Investigator Project: Development of an Integrated Process for Enhanced Mechanical Properties of Additively Manufactured CFRP using Recycled Carbon Fibers from End-of-Life Products
Apr. 2024	**National Research Foundation (NRF) Grant - Outstanding Ph.D. Student Program** **Role: Principal Investigator** Project: A Study on Material Characteristics of rCFRP with Additive Manufacturing for Sustainable Society
Oct. 2023	Living Lab Project: Social problem-solving project using technology in core areas (Daejeon · Sejong · Chungnam Regional Innovation Platform) Role: Project Leader Project: Manufacturing eco-friendly mobility components using recycled carbon fiber composites
Mar. 2023	Living Lab Project: Social problem-solving project using technology in core areas (Daejeon · Sejong · Chungnam Regional Innovation Platform) Role: Project Leader Project: For the safety of the local community, developed an electric kickboard battery protector made from recycled carbon fiber reinforced plastic
Aug. 2022	Carbon Start Up Hacker Road (Ministry of Trade, Industry and Energy & Korea Carbon Industry Promotion Agency) Role: Project Leader Project: Recycled carbon fiber undercover for electric vehicle batteries
Mar. 2022	Local Talent Integration Clubs (Daejeon · Sejong · Chungnam Regional Innovation Platform) Role: Club Leader Scope: Club management, project planning and coordination, budgeting, resource allocation, and documentation.

HONORS AND AWARDS

Mar. 2026	Best Paper Award, Korean Society for Precision Engineering Conference (KSPE Autumn 2025)
Mar. 2026	Outstanding Paper Award, Korean Society of Manufacturing Process Engineers Conference (KSMPE 2025)
Dec. 2025	**<u>President's Award, National Research Foundation of Korea (NRF)</u>** (Outstanding Graduate Student in Overseas Research, 2025 BK21 FOUR Program)
Aug. 2025	**<u>President's Doctoral Award</u>**, Kongju National University (Awarded for outstanding doctoral achievements and research excellence)
Jul. 2025	**<u>Best Paper Award</u>**, International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2025, Paper title: AI-Based Mechanical Properties Prediction of Sintering-Aging Combined Effect in Metal-Material Extrusion, Invited Journal in International Journal of Precision Engineering Green-Tech)
Jul. 2025	Outstanding Presentation Award, International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2025, Paper title: Fatigue behavior of recycled CFRP at elevated temperatures)
Nov. 2024	Best Paper Award, Korean Society for Precision Engineering Spring Conference (KSPE Autumn 2024)
Nov. 2023	Outstanding Paper Award, Korean Society for Precision Engineering Spring Conference (KSPE Autumn 2023)
Dec. 2022	Outstanding Paper Award, Korean Society of Manufacturing Process Engineers Autumn Conference (KSMPE Autumn 2022)
Aug. 2022	Government Scholarship (Chungnam Institute for Human Resources Development)
Apr. 2022	Outstanding Paper Award, Korean Society of Manufacturing Process Engineers Spring Conference (KSMPE Spring 2022)
Oct. 2021	Outstanding Paper Award, Korean Society for Precision Engineering Spring Conference (KSPE Spring 2021)
2019–2021	Corporate Scholarship (KISWEL)
Nov. 2020	Corporate Scholarship (Korea Mech Const Constructors Association)
Aug. 2017	**<u>Prime Minister's Prize</u> of Korea Youth Invention Idea Contest (Korean University Invention Association)**
Aug. 2016	Government Scholarship (Cheonan-Si Scholarship Foundation)
Aug. 2015	Government Scholarship (Cheonan-Si Scholarship Foundation)

PUBLICATIONS

1. **S. Jeon**, S. J. Park, S.K. Moon, and D. Yang*, "End-to-end processing of recycled carbon fiber composites: From waste hydrogen tanks to mechanically enhanced additive-manufactured components", ***Chemical Engineering Journal***. (SCIE, IF: 13.5, JCR<3%)
2. **S. Jeon**, S. J. Park, S.K. Moon, and D. Yang*, "Enhanced mechanical properties and reduced anisotropy of material extrusion-manufactured short carbon fibre-reinforced plastic via cold isostatic pressing", ***Virtual and Physical Prototyping***, 2025, e2499934. (SCIE, IF: 9.8, JCR<10%)
3. **S. Jeon**, J. Kim, S. J. Park, J.Y. Kim, S.K. Moon, and D. Yang*, "Temperature Dependent Mechanical Properties of End-of-Life Carbon Fiber Reinforced Plastics", ***International Journal of Precision Engineering and Manufacturing-Green Technology***, 2025. (SCIE, IF: 6.6, JCR<10%)
4. **S. Jeon**#, J. Kim#, A. Go, J.P. Choi, D. Yang* and S.K. Moon*, "AI-Based Mechanical Properties Prediction of Sintering-Aging Combined Effect in Metal-Material Extrusion", ***International Journal of Precision Engineering and Manufacturing-Green Technology***, *(Invited Paper via PRESM 2025 Best Paper Award)*. (SCIE, IF: 6.6, JCR<10%)
5. **S. Jeon**, J. Kim, and D. Yang*, "Design of Large-Scale Microwave Cavity for Uniform and Efficient Plastic Heating", ***Polymers*** 2022, 14, 541 (SCIE, IF: 5.8, Q1)
6. **S. Jeon**, Y. Kim, and D. Yang*, "Analysis of Powder Packing for Alumina Using Design of Experiment with Mixture and Vibration" ***Composites Research*** 2021, 34, 330-336
7. **S. Jeon**#, A. Go#, D. Kim, S. K. Moon*, S. J. Park*, "Intelligent process control and stimulus-responsive 4D transformation in extrusion-based 3D food printing: A critical review", ***International Journal of Bioprinting***, (SCIE, IF: 6.0).
8. J. Kim, **S. Jeon**, S. J. Park, and S.K. Moon*, "A data-efficient machine learning approach for amorphous Fe-based bulk metallic glass fabrication in powder bed fusion", ***International Journal of AI for Materials and Design***.
9. J. Lee, S. J. Park, J.P. Choi, **S. Jeon**, S.K. Moon*, "Design Complexity-Driven Optimization of Production Efficiency and Scheduling for Metal-Polymer Hybrid Structures in Additive Manufacturing", ***International Journal of Precision Engineering and Manufacturing-Green Technology***, (SCIE, IF: 6.6, JCR<10%).
10. S. J. Park, S. Cho, P.H. Nicholas Ng, J. Lee, J. Kim, **S. Jeon**, and S.K. Moon*, "Insights into the industrial applications and future directions of bulk metallic glasses via additive manufacturing", ***IJPEM*** (SCIE, IF: 3.6).
11. S. J. Park, S. Cho, **S. Jeon**, and S.K. Moon*, "A Representative Volume Element-Based Approach for Predicting Sintering Shrinkage in Metal Binder Jetting", ***International Journal of Precision Engineering and Manufacturing-Green Technology***, (SCIE, IF: 6.6, JCR<10%)

UNDER REVIEW AND SUBMITTED PAPERS (available upon request)

1. **S. Jeon**, S.K. Moon, and D. Yang*, "Achieving structural integrity in additively manufactured rCFRP via optimal fiber loading followed by cold isostatic pressing", ***Composite Structures*** *(Minor Revision)*. (JCR<5%)
2. **S. Jeon**, Y.S. Kim, Y.S. Kim, and S.K. Moon*, "Enhanced interlaminar performance of material extrusion-fabricated dual-matrix continuous carbon fiber composites", ***Polymer Testing*** *(Major Revision)*. (JCR<5%)
3. **S. Jeon**, J. Kim, J.P. Choi, D. Yang* and S.K. Moon*, "Recycling and Remanufacturing of CFRP for Additive Manufacturing and Its Applications: A Critical Review", ***International Journal of Precision Engineering and Manufacturing-Green Technology***, *(Invited Paper, Until Oct)*. (JCR<10%)
4. **S. Jeon**, S. Cho, and S.K. Moon*, "Active-learning-assisted inverse design of graded stochastic Voronoi honeycombs under an initial peak-force constraint", ***Journal of Computational Design and Engineering***, *(Under Review)*. (JCR<5%)
5. S. J. Park, **S. Jeon**, P.H. Nicholas Ng, and S.K. Moon*, "Additive manufacturing of FeCrBSiC bulk metallic glasses: Toward exploration of process parameters and trabecular-inspired hybrid structures", ***Progress in Additive Manufacturing***, *(Under Review)*.

WORKING PAPERS (**: available upon request)

1. **[Composite Recycling and Remanufacturing]** **S. Jeon**, N. Kim, J.Y. Kim*, "Closed-Loop Recycling and Upcycling of CFRP through Surface Modification" (**)
2. **[Mechanical Performance and Durability]** T. Kim, A. Go, S. Park, and **S. Jeon***, "Temperature-Dependent Mechanical Properties of Recycled CFRP Fabricated by Large-Format Additive Manufacturing" (**)
3. **[AI-Driven Materials Processing]** **S. Jeon**, J. Kim, Y. Kim*, S.K. Moon*, "Machine Learning-based Process Optimization for Tailoring Mechanical Properties of Material Extrusion-fabricated Recycled Carbon Fiber Reinforced Plastic" (**)
4. **[Advanced Manufacturing and Interface Engineering]** **S. Jeon**, Ara. Go, and S.K. Moon*, "Ultrasonic nanocrystal surface modification of 316L/17-4PH multi-material structures fabricated by material extrusion additive manufacturing and hot isostatic pressing"
5. **[Mechanical Performance and Durability]** **S. Jeon**, S.K. Moon*, "Fatigue Performance of Recycled CFRP and Feasibility Assessment for Composite Mold Fabrication"
6. **[Advanced Manufacturing and Interface Engineering]** **S. Jeon**, Zhang Qi, D. Lee, and Seung Ki Moon*, "Additive Manufacturing of Bio-Inspired Surface Lattice Design for Enhanced Interfacial Strength in Polymer-Metal Hybrid Joints".

TECHNICAL TRANSFER

1. **S. Jeon**, D. Yang*, "Optical Method for Chemical Contamination Visualization and Analysis", ***$3,550***

TECHNICAL SKILLS

Recycling and Recovery
Microwave pyrolysis of end-of-life CFRP; chemical recycling (solvolysis); carbon fiber reclamation from waste hydrogen tanks; fiber surface modification and sizing; compounding and filament extrusion (3devo Precision 350, 3devo Composer 450).

Manufacturing and Processing
Metal powder bed fusion (EOS M290); metal material extrusion (MEX); CCFRP FDM printing (Anisoprint, Markforged); IDEX FDM printing (BCN3D); compression molding; hot pressing; cold isostatic pressing (max. 3000 bar); drying and heating.

Materials Characterization
Drop weight testing (Instron); universal testing machines (Shimadzu, Instron, MTS); DSC, TGA, DMA, XRD, XPS, SEM, FT-IR, and AFM.

Data-Driven Processing and Analysis
AI-driven process optimization; machine learning and active learning; design of experiments (DOE); Python (scikit-learn, PyTorch, pandas); Minitab, Origin, MATLAB, and MS Office.

Design and Multiscale Simulation
Molecular dynamics (Materials Square); COMSOL, CATIA, SolidWorks, Ansys, and EDEM.

TEACHING EXPERIENCE

Teaching Assistant (TA) (Feb. 2020 – Aug. 2025)

- Department of Mechanical and Automotive Engineering, Kongju National University, Cheonan, Korea
  - Assistance for professors with sophomore, junior, senior and master level courses by teaching classes, managing lab sessions, holding office hours, discussion sessions and grading reports, quizzes and homework.
  - Courses include: Capstone Design, Solid Mechanics, 3D CAD, R&D Planning, Mechanical Engineering Research
- Department of Advanced Sensor Convergence Device, Daejeon · Sejong · Chungnam Regional Innovation Platform
  - Assistance for professors with freshman and sophomore level courses by teaching classes, holding office hours and discussion sessions.
  - Courses include: Sensor Engineering (131001), Micro and Nano System (131009)

Lecturer (Aug. 2025 – Dec. 2025)

- Department of Mechanical and Automotive Engineering, Kongju National University, Cheonan, Korea
  - Courses include: Mechanical Material Science and Engineering

PROFESSIONAL CERTIFICATE

Mar. 2023	Analytical Technology and Manufacturing Practice for Improving the Anti-Radiation Properties of Carbon Composites (Korea Carbon Industry Promotion Agency)
Oct. 2022	Carbon Composite Property Evaluation Technology (Korea Carbon Industry Promotion Agency)
Aug. 2022	Hot Pressing and Molding for Mass Production (Korea Carbon Industry Promotion Agency)
Jun. 2022	Carbon Composite Manufacturing Using Thermoplastics (Korea Carbon Industry Promotion Agency)
Apr. 2021	COMSOL Multiphysics Training: Fundamentals & Plasma (Altsoft)
Apr. 2021	Altair's Training: DEM Analysis and Optimization with Hyperworks (Altair)
Jan. 2020	MathWorks Training: MATLAB Fundamentals & Simulink for System and Algorithm Modeling (MathWorks)
Nov. 2019	Minitab Training: Fundamentals & DOE (Iretech)

INTERNATIONAL CONFERENCE PRESENTATIONS

1. **S. Jeon**, Zhang Qi, D. Lee, and Seung Ki Moon*, "Additive Manufacturing of Bio-Inspired Surface Lattice Design for Enhanced Interfacial Strength in Polymer-Metal Hybrid Joints", AM SMART Sep 14-16, 2026, Netherlands.

2. **S. Jeon**, J. Kim, S.K. Moon*, and D. Yang*, "Machine Learning-based Process Optimization for Tailoring Mechanical Properties of Material Extrusion-fabricated Recycled Carbon Fiber Reinforced Plastic", International Conference on Future of AM Aug 5-7, 2025, Singapore.

3. **S. Jeon**, J. Kim, A. Go, J.P. Choi, D. Yang* and S.K. Moon*, "AI-based Mechanical Properties Prediction of Sintering-aging Combined Effect in Metal-material Extrusion", International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2025), Jul 6-11, 2025, Chiang Mai, Thailand. **[Best Paper Award]**

4. **S. Jeon**, S.K. Moon, Y. Kim, and D. Yang*, "Fatigue Characteristics of Recycled Carbon Fiber Reinforced Plastic at Elevated Temperatures", International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2025), Jul 6-11, 2025, Chiang Mai, Thailand. **[Outstanding Presentation Award]**

5. J. Lee, S.K. Moon*, S. J. Park, J.P. Choi, **S. Jeon**, and J. Kim, "Design Complexity-driven Optimization of Production Efficiency and Scheduling for Metal-polymer Hybrid Structures in Additive Manufacturing", International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2025), Jul 6-11, 2025, Chiang Mai, Thailand.

6. **S. Jeon**, S.K. Moon and D. Yang*, "Mechanical properties of 17-4 PH stainless steel fabricated via metal material extrusion under sintering and aging conditions", 6th International Conference on Manufacturing Process & Technology 2025, Feb 10-13, 2025, Bali, Indonesia.

7. **S. Jeon**, S.K. Moon and D. Yang*, "Enhancing Mechanical Properties of reclaimed Carbon Fiber Reinforced Plastic Fabricated Material Extrusion via Cold Isostatic Pressing", IEEE International Conference on Industrial Engineering and Engineering Management (IEEM 2024), Dec 15-18, 2024, Bangkok, Thailand.

8. **S. Jeon**, C. Lee, and D. Yang*, "Mechanical Analysis of Recycled Carbon Fiber Composites from Waste Hydrogen Tanks", Europe-Korea Conference on Science and Technology (EKC 2023), Aug 14-18, 2023, Munich, Germany.

9. **S. Jeon**, D. Yang, and Y. Ko, "Recovering Carbon Fiber from Carbon-fiber-reinforced Polymers by Microwave Irradiation", International Symposium on Precision Engineering and Sustainable Manufacturing (PRESM 2022), Jul 20-22, 2022, Seogwipo, Republic of Korea.

10. D. Yang and **S. Jeon**, "Fabrication of Tattoo Paper-Based SERS Devices and Pesticides Sensing on Fruit Surfaces", Materials Research Society Spring Meeting & Exhibit (MRS Spring 2022), May 23-25, 2022, Honolulu, USA.

11. D. Lee, H. Lee, D. Yang, K. Lee, and **S. Jeon**, "A Study on the Locomotive Advantages of a Multi-Modular Helical Magnetic Millirobot in Curved and Narrow Blood Vessels", 2022 Joint MMM-Intermag Conference (INTERMAG), Jan 10-14, 2022, New Orleans, USA.

12. D. Yang, H. Kim, M. Ko, J. Kim, **S. Jeon**, and D. Lee, “Fabrication of Tattoo-Type Surface-Enhanced Raman Scattering (SERS) Substrate for Chemical Sensing”, The 6th International Conference on Advanced Electromaterials (ICAE 2021), Nov 9-12, 2021, Jeju, Republic of Korea.
13. **S. Jeon**, D. Yang, and D. H. Kim, “Analysis of Mixing Performance according to RPM and Particle Size of Plough for Ploughshare Mixer”, International Symposium on Precision Engineering and Sustainable Manufacturing (PRESM 2021), Jul 21-23, 2021, Jeju, Republic of Korea.
14. **S. Jeon**, J. Kim, H. Lee, and D. Yang, “Experimental Drying Characteristics of Polymer Pellets using Microwave”, International Symposium on Precision Engineering and Sustainable Manufacturing (PRESM 2020), Nov 15-18, 2020, Virtual.
15. H. Lee, D. Yang, K. Lee, and **S. Jeon**, “A Two-Dimensional Magnetic Gripper Robot Driven by a Triad of Electromagnetic Coils”, The 65th Annual Conference on Magnetism and Magnetic Materials (MMM 2020), Nov 2-6, 2020, Virtual.